# VSDBABYSOC

# MULTI-CORNER STATIC TIMING ANALYSIS

Final PVT Corner Assessment Report



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|---|---|
| **Name** | T. Murali |
| **Roll Number** | 24EG104C35 |
| **Department / Section** | ECE / C |



| | |
|---|---|
| **Design** | VSDBabySoC |
| **Technology** | SKY130 FD SC HD standard-cell paths |
| **Analysis** | Setup • Hold • WNS • TNS across 16 PVT corners |

## Assessment objective

Document the multi-corner STA results, identify the timing-limiting PVT region, show representative path-level evidence, and present a clear reproducibility structure using the supplied project evidence.

Source basis: supplied STA report, four supplied timing plots, and the grading rubric.

# 1. Assessment Objective and Evidence Map

Organized around the grading areas that affected the previous submission.

| Assessment area | Evidence included | Status |
|---|---|---|
| PVT corners | Complete 16-corner table with setup, hold, WNS and TNS values. | Included |
| STA setup | Design, technology, path group/type, library-cell evidence and a reproducibility checklist. Actual loading screenshots should be attached from the OpenSTA workspace. | Documented + checklist |
| Timing results | Setup, WNS, TNS and hold plots plus the complete numerical table. | Included |
| Evidence | Critical-path table, cell-level path sequence, recreated plots and original supplied images in the appendix. | Included |
| Interpretation | Corner trends, worst corner, critical path and timing-closure discussion. | Included |

**The supplied STA report contains timing-path output and standard-cell/library names, but it does not contain complete direct screenshots of Liberty, netlist and SDC loading. Those should be attached from the actual run if the evaluator requires execution evidence.**

# 2. STA Methodology and Reproducibility

A clear flow for repeating the same analysis.

| Step | Required evidence | Purpose |
|---|---|---|
| 1 | Design / synthesized netlist | Load VSDBabySoC into OpenSTA. |
| 2 | Corner Liberty | Load the corresponding sky130_fd_sc_hd timing library. |
| 3 | Macro libraries | Load required hard-IP timing models, if used. |
| 4 | SDC / clock | Load the constraints used for setup and hold analysis. |
| 5 | Link / update | Link the design and update timing. |
| 6 | Max-path report | Generate setup timing and identify WNS. |
| 7 | Min-path report | Generate hold timing and identify minimum hold slack. |
| 8 | All 16 corners | Repeat and record setup, hold, WNS and TNS. |

**The supplied report identifies Path Group: clk and Path Type: max and uses sky130_fd_sc_hd standard-cell timing paths. The final submission should retain the actual terminal screenshots for the loading and reporting steps.**

# 3. PVT Corner Definition

The complete 16-corner set represented in the supplied evidence.

| Group | Corners | Interpretation |
|---|---|---|
| FF | 6 corners: 100C/–40C across 1.56–1.95 V conditions | Fast-fast process group. |
| SS | 8 corners: 100C and –40C across 1.28–1.76 V conditions | Slow-slow process group. |
| TT | 2 corners: 25C/100C at 1.80 V | Typical-typical process group. |

**Coverage: 6 FF + 8 SS + 2 TT = 16 corners.**

# 4. Complete 16-Corner Timing Results

Main numerical reference table.

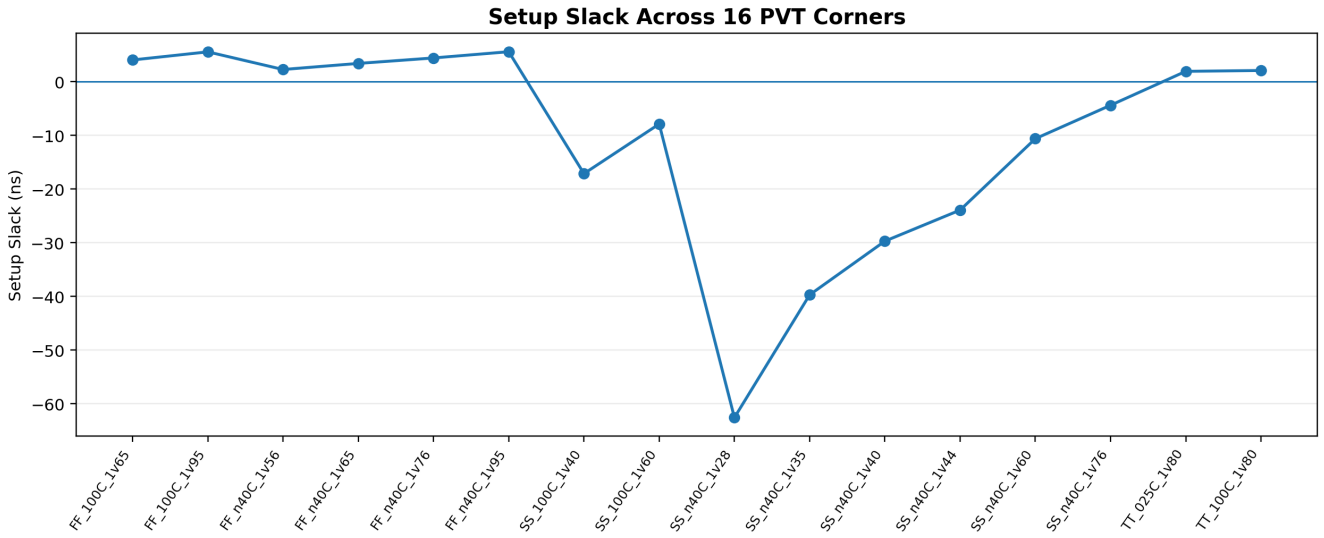
| No. | PVT Corner | Hold Min (ns) | Setup Max (ns) | WNS (ns) | TNS (ns) | Status |
|---|---|---|---|---|---|---|
| 1 | FF_100C_1v65 | 0.249 | 4.055 | 0.000 | 0.000 | MET |
| 2 | FF_100C_1v95 | 0.196 | 5.567 | 0.000 | 0.000 | MET |
| 3 | FF_n40C_1v56 | 0.292 | 2.278 | 0.000 | 0.000 | MET |
| 4 | FF_n40C_1v65 | 0.255 | 3.411 | 0.000 | 0.000 | MET |
| 5 | FF_n40C_1v76 | 0.224 | 4.430 | 0.000 | 0.000 | MET |
| 6 | FF_n40C_1v95 | 0.187 | 5.592 | 0.000 | 0.000 | MET |
| 7 | SS_100C_1v40 | 0.905 | -17.122 | -17.122 | -2,599.514 | VIOLATED |
| 8 | SS_100C_1v60 | 0.642 | -7.874 | -7.874 | -892.705 | VIOLATED |
| 9 | SS_n40C_1v28 | 1.830 | -62.563 | -62.563 | -13,957.588 | VIOLATED |
| 10 | SS_n40C_1v35 | 1.348 | -39.697 | -39.697 | -7,993.793 | VIOLATED |
| 11 | SS_n40C_1v40 | 1.125 | -29.694 | -29.694 | -5,466.484 | VIOLATED |
| 12 | SS_n40C_1v44 | 0.991 | -23.908 | -23.908 | -4,107.187 | VIOLATED |
| 13 | SS_n40C_1v60 | 0.663 | -10.609 | -10.609 | -1,355.797 | VIOLATED |
| 14 | SS_n40C_1v76 | 0.504 | -4.382 | -4.382 | -367.928 | VIOLATED |
| 15 | TT_025C_1v80 | 0.310 | 1.939 | 0.000 | 0.000 | MET |
| 16 | TT_100C_1v80 | 0.314 | 2.094 | 0.000 | 0.000 | MET |

Eight SS corners are setup-violating in the supplied comparison. All FF and TT entries have positive setup slack, while all displayed hold minimum-slack values remain positive.

# 5. Setup Timing Analysis

Maximum setup slack across the 16 supplied PVT corners.

## Figure 1: Setup Timing Analysis




Plot recreated directly from the supplied numerical timing values.

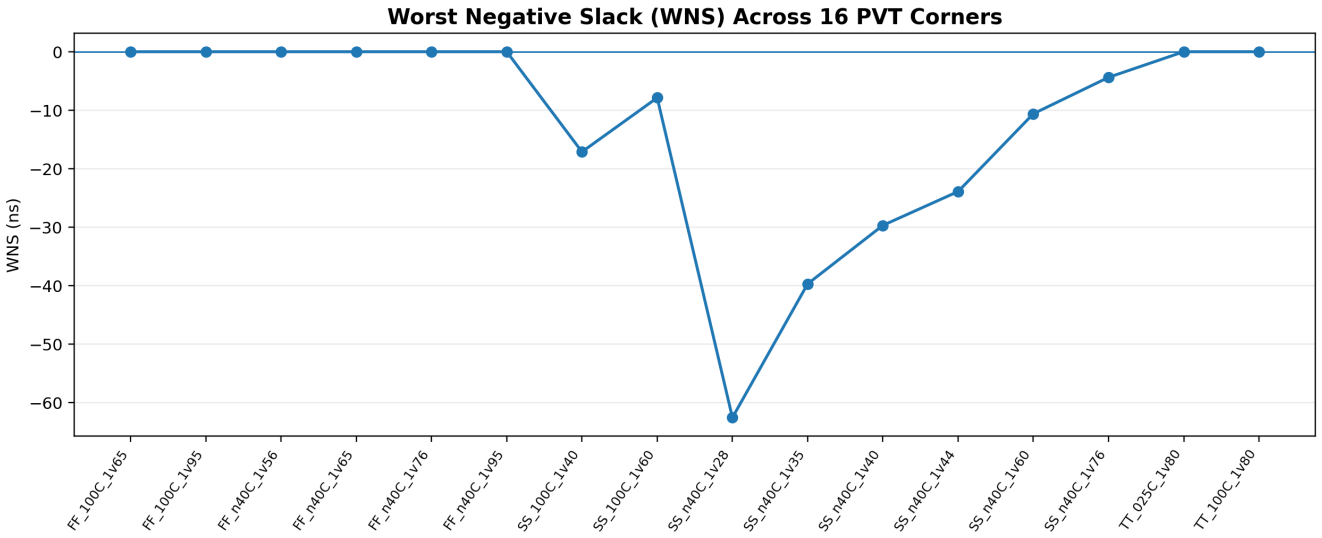
The FF group remains positive. All eight SS corners are negative, with SS_n40C_1v28 reaching –62.563 ns. The two TT corners return to positive setup slack.

**Finding: the dominant timing concern is setup in the SS corner group.**

# 6. Worst Negative Slack (WNS)

Worst setup slack at each PVT corner.

## Figure 2: Worst Negative Slack (WNS)



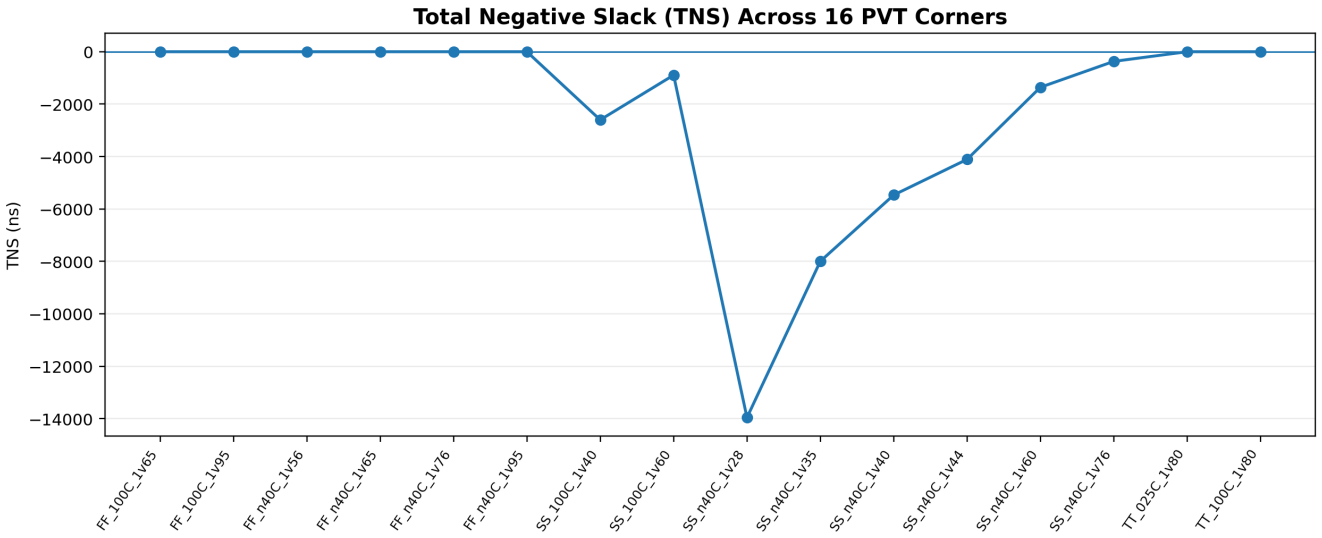

Plot recreated directly from the supplied numerical timing values.

The worst WNS is –62.563 ns at SS_n40C_1v28, followed by –39.697 ns at SS_n40C_1v35 and –29.694 ns at SS_n40C_1v40.

# 7. Total Negative Slack (TNS)

Accumulated negative setup slack.

## Figure 3: Total Negative Slack (TNS)



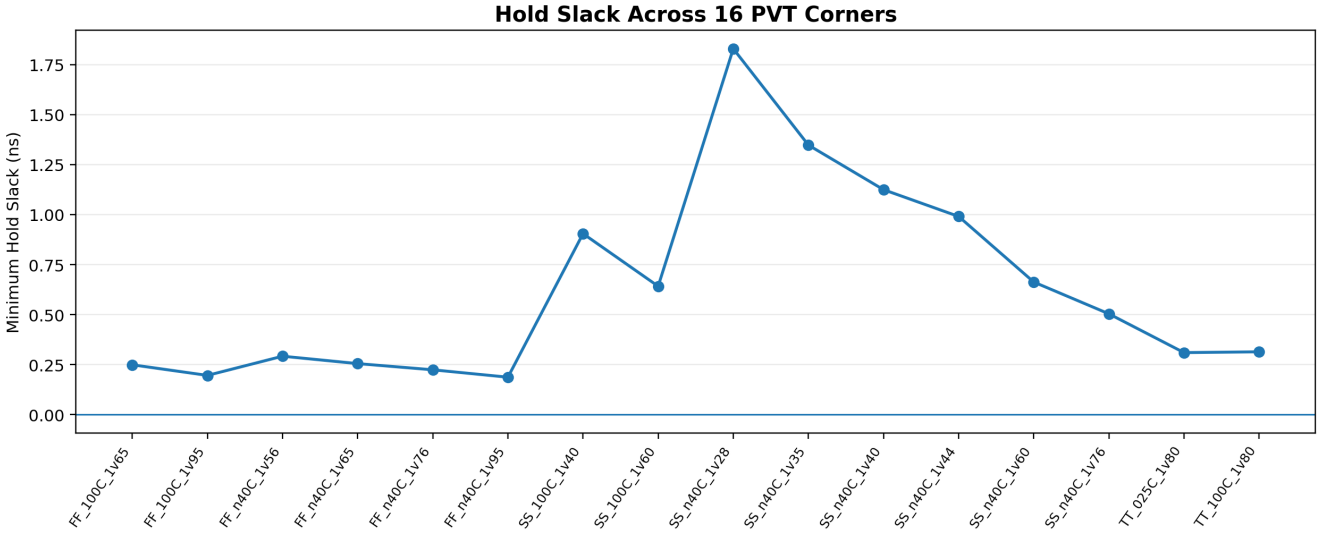

Plot recreated directly from the supplied numerical timing values.

TNS is zero for the MET corners and negative for every SS corner. The most negative TNS is –13,957.588 ns at SS_n40C_1v28.

# 8. Hold Timing Analysis

Minimum hold slack across the 16 supplied PVT corners.

## Figure 4: Hold Timing Analysis




Plot recreated directly from the supplied numerical timing values.

All displayed minimum hold-slack values are positive. The smallest value is 0.187 ns at FF_n40C_1v95; the largest is 1.830 ns at SS_n40C_1v28.

**Finding: the supplied evidence does not show a negative hold-slack point.**

# 9. Critical Path Evidence

Path-level values from the supplied STA report.

| Parameter | Supplied value |
|---|---|
| Startpoint | core/CPU_is_addi_a3_reg |
| Endpoint | core/CPU_Xreg_value_a4_reg[27][30] |
| Path group | clk |
| Path type | max |
| Critical corner | SS_n40C_1v28 |
| Data arrival time | 71.70 ns |
| Data required time | 9.14 ns |
| Slack | –62.56 ns |
| WNS / setup value | –62.563 ns |
| TNS | –13,957.588 ns |

The path-level values provide a direct cross-check of the plotted setup result. The large data-arrival time relative to the required time is the reason the reported slack is negative at this corner.

**Critical path: core/CPU_is_addi_a3_reg → combinational logic → core/CPU_Xreg_value_a4_reg[27][30].**

# 10. Critical Path Cell-Level Evidence

Standard-cell instances visible in the supplied path report.

| Path instance | Cell type |
|---|---|
| core/CPU_is_addi_a3_reg/CLK | sky130_fd_sc_hd__dfxtp_1 |
| core/CPU_is_addi_a3_reg/Q | sky130_fd_sc_hd__dfxtp_1 |
| core/U474/Y | sky130_fd_sc_hd__nor2_1 |
| core/U476/Y | sky130_fd_sc_hd__nand2_1 |
| core/U9/Y | sky130_fd_sc_hd__inv_2 |
| core/U479/X | sky130_fd_sc_hd__xor2_1 |
| core/U569/Y | sky130_fd_sc_hd__a21oi_1 |
| core/U571/Y | sky130_fd_sc_hd__o21ai_1 |
| core/U574/Y | sky130_fd_sc_hd__a21oi_1 |
| core/U576/Y | sky130_fd_sc_hd__o21ai_1 |
| core/U581/Y | sky130_fd_sc_hd__a21oi_1 |
| core/U583/Y | sky130_fd_sc_hd__o21ai_1 |
| core/U587/Y | sky130_fd_sc_hd__a21oi_1 |
| core/U589/Y | sky130_fd_sc_hd__o21ai_1 |
| core/U344/Y | sky130_fd_sc_hd__a21oi_1 |
| core/U213/Y | sky130_fd_sc_hd__o21ai_1 |
| core/U343/Y | sky130_fd_sc_hd__a21oi_1 |
| core/U210/Y | sky130_fd_sc_hd__o21ai_1 |
| core/U342/Y | sky130_fd_sc_hd__a21oi_1 |
| core/U209/Y | sky130_fd_sc_hd__o21ai_1 |

| Path instance | Cell type |
|---|---|
| core/U341/Y | sky130_fd_sc_hd__a21oi_1 |
| core/U215/Y | sky130_fd_sc_hd__a21o_1 |
| core/U96/X | sky130_fd_sc_hd__a21o_1 |
| core/U942/COUT | sky130_fd_sc_hd__fa_1 |
| core/U94/X | sky130_fd_sc_hd__a21o_1 |
| core/U905/COUT | sky130_fd_sc_hd__fa_1 |
| core/U887/COUT | sky130_fd_sc_hd__fa_1 |
| core/U1370/COUT | sky130_fd_sc_hd__fa_1 |
| core/U36/COUT | sky130_fd_sc_hd__fa_2 |
| core/U87/Y | sky130_fd_sc_hd__clkinv_1 |
| core/U613/Y | sky130_fd_sc_hd__o21ai_1 |
| core/U352/COUT | sky130_fd_sc_hd__fa_1 |
| core/U348/COUT | sky130_fd_sc_hd__fa_1 |
| core/U35/COUT | sky130_fd_sc_hd__fa_1 |
| core/U33/COUT | sky130_fd_sc_hd__fa_1 |
| core/U81/X | sky130_fd_sc_hd__a21o_1 |
| core/U347/COUT | sky130_fd_sc_hd__fa_1 |

| Path instance | Cell type |
|---|---|
| core/U351/SUM | sky130_fd_sc_hd__fa_1 |
| core/U3/Y | sky130_fd_sc_hd__nand2_1 |
| core/U696/Y | sky130_fd_sc_hd__o22ai_1 |

The supplied path contains repeated NAND/NOR/AOI/OAI/XOR/inverter and full-adder stages. This is direct path evidence; the source report does not by itself prove which individual cell is the root cause.

# 11. Timing-Closure Review Plan

Actions to take from the measured results.

| Priority | Review action | Reason | Verification |
|---|---|---|---|
| 1 | Inspect SS_n40C_1v28 first. | Worst setup/WNS/TNS. | Rerun critical max path. |
| 2 | Trace the reported critical path. | 71.70 ns arrival time. | Compare path delay after any change. |
| 3 | Compare neighboring SS corners. | Several remain strongly negative. | Check whether the same path remains critical. |
| 4 | Protect hold margin. | All supplied hold values are positive. | Run min-path STA after setup changes. |
| 5 | Rerun all 16 corners. | Closure must be demonstrated across PVT. | Record updated WNS/TNS/setup/hold. |

**The supplied evidence establishes the timing problem but does not document a completed optimization experiment. Therefore, this report does not claim an optimization has already fixed the violations.**

# 12. Final Evidence Checklist

What to keep with the report when submitting.

| Evidence item | In this report | Recommended actual-run attachment |
|---|---|---|
| 16-corner PVT table | Yes | — |
| Setup/WNS/TNS/Hold graphs | Yes | — |
| Critical setup path | Yes | Actual OpenSTA terminal screenshot |
| Liberty loading | Method described | Screenshot of actual corner .lib loading |
| Netlist/linking | Method described | Screenshot of actual netlist/link command |
| SDC/clock | Method described | Screenshot of actual SDC/clock loading |
| Per-corner execution | 16 results documented | Representative FF/SS/TT terminal runs |
| Hold evidence | Positive values documented | Actual min-path screenshot if available |

This checklist distinguishes supplied evidence from evidence that should come directly from the actual OpenSTA workspace. No execution screenshot is invented.

# 13. Conclusion

Final assessment statement.

The 16-corner STA comparison shows a clear process-dependent timing pattern. All six FF corners and both TT corners have positive setup slack, while all eight SS corners have negative setup slack. The worst setup/WNS result is SS_n40C_1v28 at –62.563 ns, with TNS of –13,957.588 ns.

The detailed STA path at SS_n40C_1v28 reports 71.70 ns data arrival time and 9.14 ns data required time, producing approximately –62.56 ns slack. The path starts at core/CPU_is_addi_a3_reg and ends at core/CPU_Xreg_value_a4_reg[27][30].

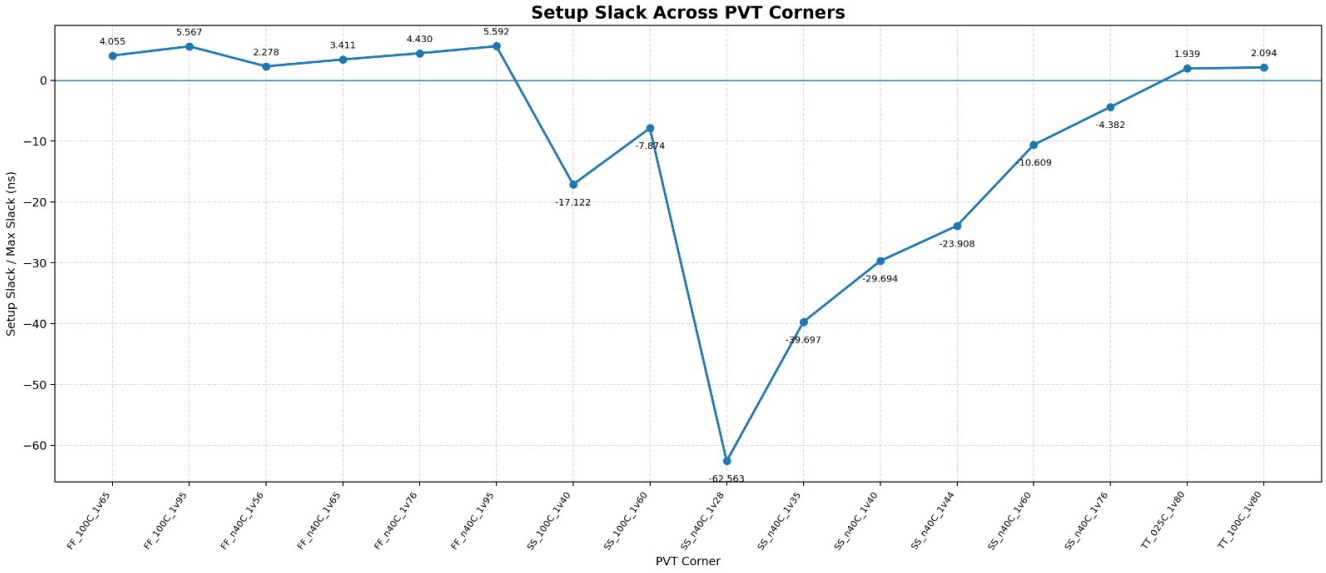
The supplied hold plot remains positive at all 16 corners, with a minimum displayed hold slack of 0.187 ns. Within the provided evidence, the principal timing-closure concern is setup timing in the SS group.

**Final engineering interpretation: the SS corner group should be the starting point for timing-closure work. Any implementation change should be validated by repeating max-path and min-path STA across the complete 16-corner set and recording updated WNS, TNS, setup and hold values.**

Source note: numerical values and path-level information are based on the supplied STA report and uploaded timing plots. The report structure has been expanded for clarity and assessment coverage without inventing execution evidence.

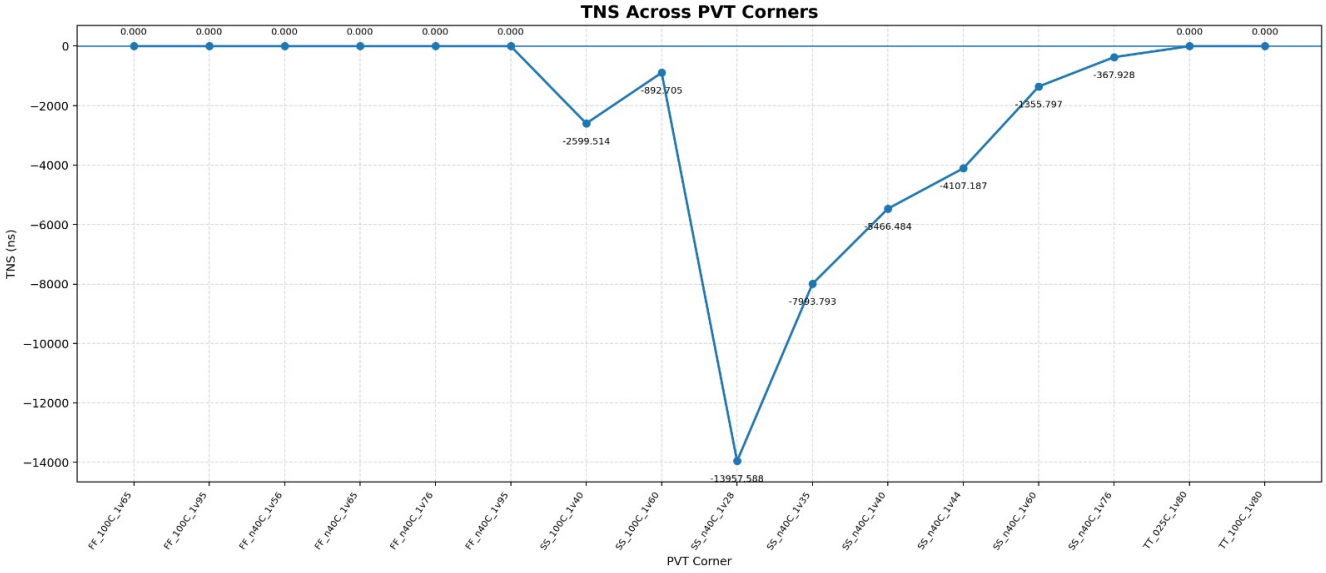
# Appendix A — Supplied Setup Evidence

Original uploaded setup-slack evidence.



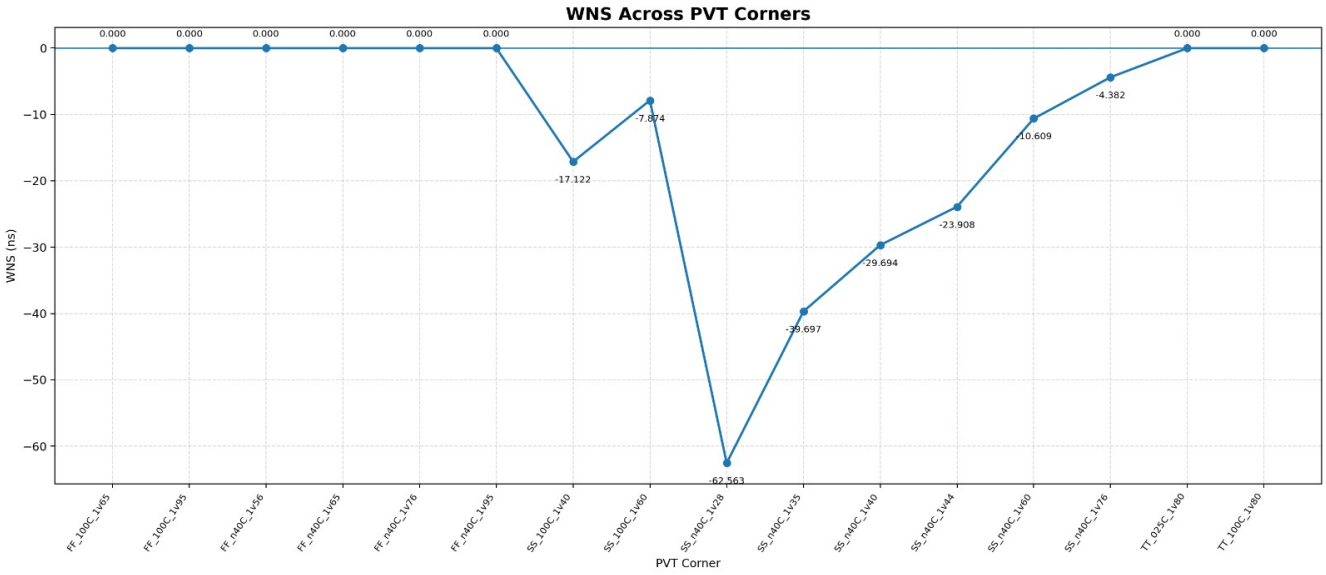
# Appendix B — Supplied TNS Evidence

Original uploaded TNS evidence.



# Appendix C — Supplied WNS Evidence

Original uploaded WNS evidence.



# Appendix D — Supplied Hold Evidence

Original uploaded hold-slack evidence.

